

YANKTON GURNEY, SD, USA

WMO: 726525

Lat: 42.917N

Lon: 97.383W

Elev: 357

StdP: 97.11

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 94911

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-22.2	-19.1	-27.6	0.3	-21.2	-23.9	0.5	-19.2	16.3	-11.0	14.9	-7.8	4.1	310	0.613

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.1	33.8	23.4	32.3	23.1	30.9	22.5	25.4	31.2	24.4	29.8	23.5	28.8	6.5	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
23.7	19.3	29.1	22.6	18.1	28.0	22.0	17.4	27.3	79.9	30.9	75.4	30.0	71.7	28.7	28.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 13.3	11.7	10.4	-26.1	36.7	3.2	1.9	-28.3	38.1	-30.2	39.2	-32.0	40.3	-34.3	41.7		
(5)			-26.4	27.1	2.8	1.0	-28.4	27.8	-30.1	28.4	-31.7	29.0	-33.7	29.7		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	9.2	-6.3	-4.2	2.3	8.9	15.4	21.1	23.9	22.2	18.1	10.4	2.7	-4.3	(6)
(7)		DBStd	11.94	7.17	7.17	7.16	5.66	4.79	3.92	3.43	3.10	4.67	5.08	6.18	6.49	(7)
(8)		HDD10.0	1984	505	399	254	84	9	0	0	0	2	57	230	443	(8)
(9)		HDD18.3	3804	763	632	498	287	117	18	2	5	61	249	470	702	(9)
(10)		CDD10.0	1709	0	1	15	51	176	333	431	379	244	70	9	0	(10)
(11)		CDD18.3	488	0	0	1	3	26	101	174	126	53	4	0	0	(11)
(12)		CDH23.3	4824	0	0	10	61	297	998	1809	1076	519	52	2	0	(12)
(13)		CDH26.7	1782	0	0	1	16	92	363	757	369	171	13	0	0	(13)
(14)	Wind	WSAvg	4.6	4.9	5.0	5.2	5.5	4.9	4.3	3.8	3.6	4.2	4.7	4.7	4.7	(14)
(15)	Precipitation	PrecAvg	590	10	12	42	56	90	96	80	74	61	39	25	14	(15)
(16)		PrecMax	899	33	46	133	180	297	178	225	199	189	140	107	50	(16)
(17)		PrecMin	371	0	0	0	10	18	24	6	7	17	2	0	1	(17)
(18)		PrecStd	133	10	11	33	36	62	42	53	47	40	37	26	11	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	12.7	17.1	24.1	28.9	32.5	35.1	37.6	35.2	32.9	27.9	22.5	12.9	(19)
(20)			MCWB	6.3	9.6	13.2	17.0	19.8	23.4	23.8	23.7	21.8	16.7	12.6	6.8	(20)
(21)		2%	DB	7.9	12.2	20.3	24.3	28.9	32.5	34.1	32.4	31.1	24.0	17.9	9.1	(21)
(22)			MCWB	3.5	7.0	12.3	14.4	18.5	22.5	24.1	23.7	21.6	15.5	11.1	5.0	(22)
(23)		5%	DB	5.8	8.7	16.3	21.4	27.1	30.9	32.6	30.9	28.5	22.1	14.9	7.0	(23)
(24)			MCWB	2.3	4.8	10.1	13.2	17.7	21.4	24.1	23.2	20.4	14.6	8.7	3.2	(24)
(25)		10%	DB	2.9	5.9	12.7	18.0	23.9	28.7	31.0	28.3	26.4	18.9	12.1	3.9	(25)
(26)			MCWB	0.7	2.5	7.8	11.1	16.3	20.6	23.5	22.0	18.7	12.9	7.4	1.3	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	6.6	10.4	15.7	18.0	21.9	25.4	27.0	26.2	24.1	19.1	14.5	8.2	(27)
(28)			MCD B	12.5	15.3	20.5	26.7	28.7	32.2	32.6	31.8	29.3	25.2	19.4	11.6	(28)
(29)		2%	WB	4.0	7.1	13.1	15.7	20.1	23.7	25.7	24.9	22.8	17.1	11.7	5.1	(29)
(30)			MCD B	7.7	11.8	18.9	22.9	26.8	30.0	31.5	30.2	28.3	22.0	17.4	9.2	(30)
(31)		5%	WB	2.4	4.6	10.7	13.7	18.7	22.6	24.9	23.9	21.6	15.4	9.4	3.2	(31)
(32)			MCD B	5.3	8.4	16.9	20.2	25.2	28.9	30.5	28.9	27.1	20.1	14.7	6.3	(32)
(33)		10%	WB	0.9	2.6	7.9	11.9	17.2	21.5	23.9	22.9	20.2	13.5	7.3	1.6	(33)
(34)			MCD B	3.1	5.5	12.6	17.4	22.9	27.4	29.4	27.4	25.0	18.6	11.4	3.9	(34)
(35)	Mean Daily Temperature Range	MDBR	9.6	10.2	10.9	11.7	11.3	11.1	11.1	11.0	12.2	11.7	10.8	9.1	(35)	
(36)		5% DB	MCD BR	13.4	14.9	16.6	16.9	14.8	13.5	12.8	13.3	14.4	16.1	16.0	13.1	(36)
(37)			MCWBR	9.7	10.2	10.2	9.1	7.4	6.3	5.8	6.6	7.1	9.0	9.9	9.3	(37)
(38)		5% WB	MCD BR	12.7	14.2	15.9	14.8	13.1	11.9	11.2	11.3	12.3	13.1	15.4	12.4	(38)
(39)			MCWBR	9.5	9.9	9.8	8.6	7.4	6.5	6.2	6.5	6.9	8.3	10.0	9.0	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.275	0.291	0.319	0.376	0.394	0.404	0.409	0.411	0.360	0.324	0.289	0.275	(40)	
(41)		taud	2.447	2.432	2.423	2.272	2.305	2.331	2.322	2.304	2.415	2.491	2.504	2.461	(41)	
(42)		Ebn at Noon	874	916	926	889	880	869	860	846	870	864	851	840	(42)	
(43)		Edh at Noon	73	88	102	128	128	125	125	123	102	83	69	66	(43)	
(44)	All-Sky Solar Radiation	RadAvg	1.94	2.95	3.88	4.92	5.75	6.45	6.70	5.59	4.49	3.08	2.14	1.59	(44)	
(45)		RadStd	0.16	0.21	0.31	0.38	0.42	0.45	0.29	0.42	0.33	0.36	0.16	0.14	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (30 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page